

EAN-Target Tracking

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Web: sightlineintelligence.com

[Contact Us](#)

Sales: +1 (541) 716-5137

Support: +1 (503) 616-3063

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 **CAUTION:** Alerts to a condition or improper procedure that can result in equipment or component damage if not avoided.

 **IMPORTANT:** Identifies information that is critical to successful setup, configuration, or operation of the system.

 Provides supplementary information that aids in the use or understanding of the system or subject matter but is not critical to operation.



1 Overview

This document provides guidelines for optimal target tracking and scene tracking performance, as well as a list of commands used for tracking.

Object tracking uses frame-to-frame transformation in conjunction with the estimated velocity of the object and previous position to estimate the position of the object in the current frame. Correlation based template matching long-term features is used to locate the object position in the current frame.

Use of the registration transformation improves tracking speed and robustness by allowing the correlation search gate to be small. This means that the object is only sought in a narrow zone around where it is expected to be located.

The registration template also permits the estimation of object velocity (in camera pixel coordinates). This estimate is used to reacquire lost tracking lock when an object is temporarily obscured.

Tracking scenarios are an exercise in judgment and understanding the tracking software limitations. What works best in one tracking case may not be ideal in another. The best way to learn and get better at target tracking is through experimentation.

In cases where the tracker is not tracking effectively, send Sightline [Support](#) a copy of the original video (without stabilization or overlays if possible) for review. This will help to improve and refine the tracking algorithms for future updates.

1.1 Additional Support Documentation

The [Panel Plus](#) User Guide is accessible from the Help menu within the application.

The Interface Command and Control (IDD) describes the native communications protocol.

Additional Engineering Application Notes (EANs) and ICDs are available on the [Product Support Documentation](#) page in the [Sightline Knowledge Base](#).

1.2 Sightline Software Requirements

IMPORTANT: The Panel Plus software version should match the firmware version running on the OEM. Firmware and Panel Plus software versions are available on the [Software Releases](#) page.

Table 1: Sightline OEM Software Requirements

OEMs	Initial Software Release	Provisions
4110-OEM	3.11.x	--
1710-OEM	3.9.x	--
4100-OEM	3.7.x	Three channel processing requires 3.9.x and above.
1750-OEM	3.6.x	Dual channel processing and Snap All Cams requires 3.7.x and above.
4000-OEM	3.00.xx	AI-Assist and AI-Detect are not supported. Blending requires 3.3.x and above.
Legacy OEMs are supported up to 3.5.x		
1500-OEM	2.20.xx	--
3000-OEM	2.22.xx	Rev B6 requires updated firmware to accommodate the change in memory.

1.3 Software Feature Version Requirements

Software feature availability is determined by software version. Refer to the table below for the minimum software version that supports each feature described in this document. For questions or upgrades, contact [Sales](#).

**Table 2: Software Feature Version Requirements**

Tracking Features	Initial Software Release
Vehicle	2.22.xx
Stationary	
Scene	
No Registration	
Static	
Acquisition Assist	
Intelligent Assist	
Drone	2.24.xx
Person	3.01.xx
Precision Acquisition	3.8.x
Twenty Tracks	3.9.x
AI Assist: AiTR, AI Detection, Detection, Tracking Base	3.10.x

2 Target Tracking Best Practices

It is important to configure the system for the type of tracking that fits the conditions. Setting the size of the track box is important. It encompasses the area within the frame that the tracker is constantly attempting to track. This is only relevant to the Vehicle, Stationary, No Registration and Drone tracking modes (see [Tracking Modes](#) for descriptions).

The optimal size of the box should encompass the entire object being tracked. A slight (~10%) region around the object is optimal to provide a good foreground/background difference.

In certain situations, this may not always be possible. Users should focus on including features with high contrast within the track box. These include corners (for example windows in a car) or other edges.

Once a track has been initiated and the tracker is following the target, it is possible to nudge the track box to get better coverage of the entire target. It is also possible to resize the existing target. The tracker uses the existing target position and retrain a model on the new adjusted location and size.

Nudging and resizing the target is preferable to re-designating the target due to latency (user input) and pixel coordinates being received by the hardware. When nudging or resizing, ensure the object being tracked is completely visible to avoid retraining the model on something else.

Processing at a constant frame rate is important for tracking performance. The display processing load can be reduced by:

- setting a display frame step or capture frame step (see [EAN-Performance-and-Latency](#))
- turning off enhancement
- choosing a faster encoder (MPEG4 for 1500-OEM board) and smaller output frame size
- reducing registration search space (set Maximum Zoom to 0 and if there is no rotation, set Maximum Rotation to 0)



The Show Performance Graph in Panel Plus shows the time used by the processing steps. From the main menu View » Show Performance Graphs.



3 Tracking Modes

The following tracking modes are supported in Sightline software.

Vehicle: Used for tracking any type of moving object. The camera may be stationary or moving in this instance.

Stationary: Used for tracking a fixed object such as a house. The camera may be stationary or moving in this instance. If the object is sometimes stationary and sometimes moving, use Vehicle or Drone mode.

Scene: Also known as Scene Tracking. Keeps the entire scene in the frame. The camera platform is moving, and no single object is selected to track. Although it is a mode exposed by the tracker, it can also be implemented by using the scene translation results directly. Sightline recommends using the scene translation results directly for a more tightly integrated solution (see FAQ below for details).

No Registration (Deprecated in 3.7): Used to track moving or non-moving objects when it is difficult to accurately estimate frame-to-frame registration. For example, when attempting to track an airplane against a blue sky.

ⓘ IMPORTANT: No registration mode should only be used when necessary as it can result in additional tracking failures in cluttered environments. The advancements in Vehicle and Drone modes within the 2.25.xx and 3.00.xx software render this mode less effective compared to other tracking modes.

Static: Not an object tracking mode. It is used to specify specific camera coordinates for temperature data reporting, for more details see the [EAN-Infrared-Temperature](#).

Drone (Software 2.24.xx and above): Optimized for fast and inconsistently moving objects such as quadcopters. It can handle objects that quickly change in size and appearance. Designed to work either as part of a ground based or aerial system.

3.1 Unique Tracks

Software 3.11.x and above.

Unique Track Modes is in the *Advanced Tracking* dialog window (*Track tab* » *Advanced tracking settings*).

Unique Track Modes enhance flexibility when tracking multiple targets in the same scene.

When this feature is enabled, tracks can be initiated using different track modes at the same time. For example, when tracking both a person and a vehicle in a single scene, the person can be tracked using the Person mode while the vehicle is tracked using the Vehicle mode.

New tracks use the track mode that is currently active. If the active track mode is changed, only tracks initiated after the change use the new mode. Existing tracks retain the mode used when they were created

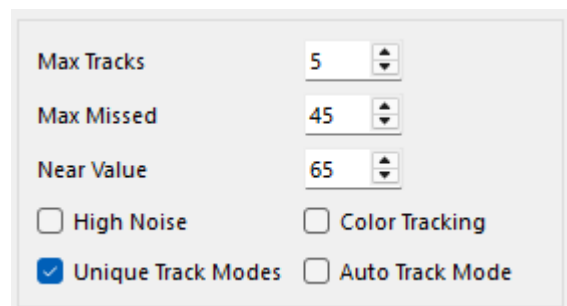


Figure 1: Unique Tracks Enabled in Panel Plus



3.2 Auto Mode

Software 3.11.x and above.

Auto Track Mode is in the *Advanced Tracking* dialog window (*Track tab* » *Advanced tracking settings*).

The *Auto Track Mode* feature automatically selects a tracking mode based on the classification of the tracked object.

When using this mode, the tracked object is passed to a classifier that will automatically set the tracking mode to either *Drone*, *Vehicle*, or *Person* based on the classification of the object.

Objects classified as a maritime vessel use the *Vehicle* track mode.

Unique Tracks must be enabled to use *Auto Track Mode*. Unlike *Unique Tracks*, enabling *Auto Track Mode* will classify all existing tracks and update them to the appropriate tracking mode.

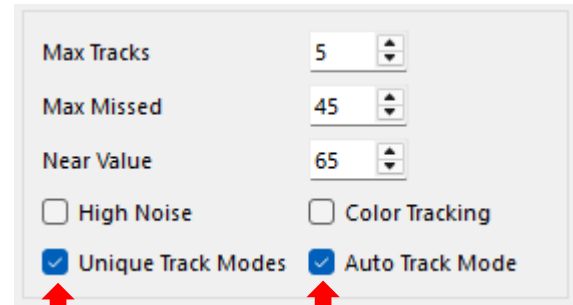


Figure 2: Auto Track Mode Enabled in Panel Plus

4 Track Box Selection

The diagram below shows six examples of attempts to initialize a track. In all cases, the object being tracked is represented by the dark rectangle. The tracking box is depicted in red.

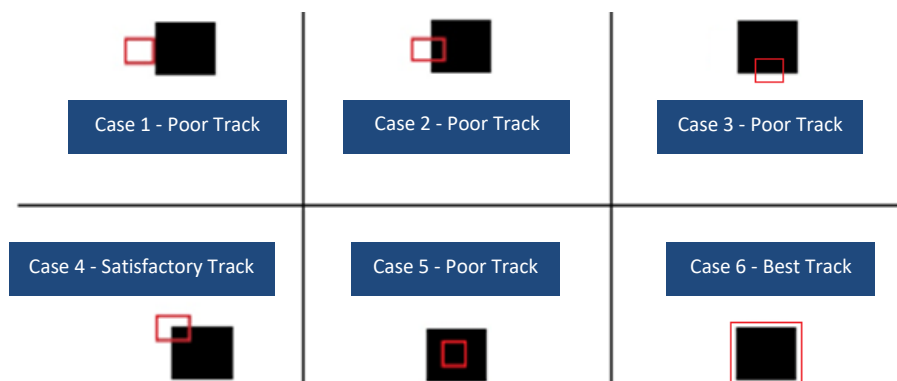


Figure 3: Designating Targets Example

Case 1: The tracking was not initialized on the object. The tracker stays on the edge of the object. If there are other good features in the background, the tracker may follow them instead of the tracked object. This often happens when there is a delay in communication to the tracker or latency in the video because the object moves from the time the user sees it on the display to the time the track command reaches the tracker.

Case 2 and Case 3: A slightly better initialization of track. Notice that the only feature to track is a single edge. The tracker will likely drift up and down (or left and right) along that edge resulting in an erratic track.



Case 4: Better than Case 2 and Case 3. There are two edges (a corner) for the tracker to follow. Tracking should be good if the background has no stronger features.

Case 5: Since there are no visible features for the tracker to work with. The tracker will either fail to initialize altogether, or it will drift to an edge or corner of the rectangle.

Case 6: Optimally, the tracking box is sized to encompass the entire object that is being tracked. This gives the tracker the most features to work with, and a minimum number of features from the background.

4.1 Vehicle Track Mode Examples

The examples shown in this section are real world examples.

In this example the car is not fully enclosed by the tracking box, but there are still good features with high contrast. This is not optimal but still works well.



Figure 4: Tracking Box too Small

In this example the tracking box is too large and encompasses more than the vehicle, but the tracking area contains enough good features and still tracks well.



Figure 5: Tracking Box too Large

In this example the tracking box is sized for optimal tracking performance. It includes the object and a small amount of background features, allowing the track to persist through occlusions and background changes.



Figure 6: Tacking Box Optimal



4.2 Person Track Mode Examples

Software 3.01.xx and above.

Many of the principles discussed in track box selection for Vehicle mode apply for Person track mode. However, when tracking people additional considerations should be considered.

The example in **Figure 7** shows a square track box. The square track box includes a lot of background and is more likely to fail over time. This example will track well for a few minutes then slide off the person.

The example in **Figure 8** shows a track box sized for optimal performance with minimal background. It is placed on the upper half of the body with a rectangular shape. This track works well through background changes, quick changes in velocity, and if the person bends over and stands up.



Figure 7: Person Tracking Box Too Large



Figure 8: Person Tracking Box Optimal

4.3 Acquisition Assist

Software 2.22.xx and above.

Acquisition Assist is designed to help users select an optimal size and location for track boxes. When this mode is enabled through the [SetTrackingParameters \(0x0C\)](#) message, an algorithm runs on track-start that will search an area around the initial track location for the optimal track box size and location. The size of the search region is based on the track box size setting, so it is still important to set the size and location. This algorithm is designed to be set close and still get good tracking results.

Tracking moving objects from a moving platform creates challenging issues. Over time, the optimal track box at initialization may not last. To resolve this, manually re-designate the track, nudge the track, or adjust the track size. Another option is to use [ModifyTrackIndex \(0x17\)](#) message and send the *Reinitialize* command. This will rerun the acquisition assist logic. If successful, it will replace the current track box with the new optimal track box. Another option is to use intelligent assist to automatically readjust under certain conditions.



4.3.1 Acquisition Assist On

In this example the track box is centered on the car and contains enough background for optimal tracking.



Figure 9: Acquisition Assist On

4.3.2 Acquisition Assist Off

In this example the track box is centered but contains too much background content and will not track as well as the example shown in [Figure 10](#).



Figure 10: Acquisition Assist Off

4.3.3 Resizing with Acquisition Assist Enabled

Software 3.00.xx and above.

There are cases when running acquisition assist will not result in an optimal track box size. Once you are tracking if you send the [SetTrackingParameters \(0x0C\)](#) message to adjust the size this will result in acquisition assist logic running again and you may still not end up with an optimal track box size. Alternatively, once a track is started use the [ModifyTrackIndex \(0x17\)](#) message with flags 8 or 9 to adjust the track size with or without re-running acquisition assist. A common use case is to start the track with acquisition assist enabled to get a good initial estimate and then adjust with the [ModifyTrackIndex \(0x17\)](#) message without acquisition assist to get an optimal track box.

4.4 Intelligent Assist

Software 2.22.xx and above.

Intelligent assist is designed to automatically reinitialize targets when changes are detected in the target appearance. These changes are currently limited to turning objects in Vehicle mode and changing sizes in Drone mode. This will be expanded to cover other cases in the future.

Enabling this mode is designed to be low risk, instead of swapping out the old track for the new one it migrates to the new track over time.



Intelligent assist requires acquisition assist to be enabled since it uses the acquisition assist algorithm to estimate the new optimal track box size and position.

Drone tracking can handle quickly changing target sizes. In most cases it is recommended to use intelligent assist to optimize the track box size while in Drone mode. There are some cases where the background clutter is high, which could negatively impact performance.

4.5 AI Track Assist

Software 3.10.xx and above.

AI Track Assist is an enhanced tracking mode that combines standard tracking functionality with AI Object Detection. This integration improves overall tracking performance while preserving standard tracking features.

Before using AI track assist, ensure that AI full frame object detection is enabled (see [EAN-Detection-Modes](#)) and that the AI is correctly detecting the object of interest. Once AI full frame object detection is active and identifying the object, AI track assist can be enabled to improve both detection and tracking performance. To enable *AI Assist*, check the box in the *Manual* tab.

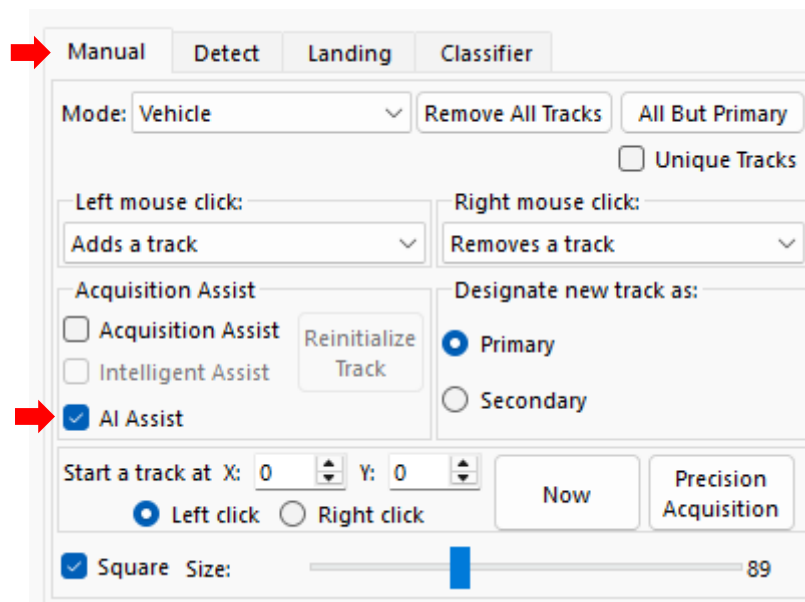


Figure 11: Enable AI Track Assist

4.5.1 AI Track Assist Indicator

To verify that AI is assisting the track, an overlay indicator can be enabled. This option is available in the *Overlay Settings* menu.

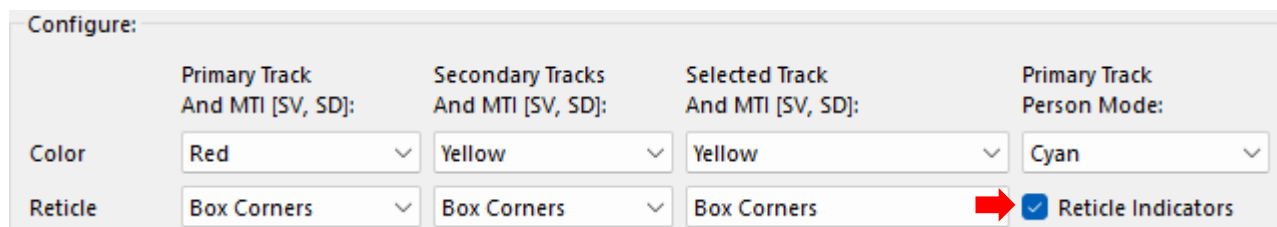


Figure 12: Enable AI Track Assist Indicator



When AI full frame object detection is actively supporting the track, an upside-down triangle appears at the top of the track box. If this indicator is visible, AI is actively assisting with tracking. If the triangle is not present, AI is not assisting the track. This can occur if the object is lost, the AI cannot detect the object, or AI is not enabled.



Figure 13: AI Assist Indicator (Triangle)

4.6 Color Tracking

Software 3.11.xx and above.

Color tracking has been added to the Sightline firmware, enabling the tracker to use color information in addition to grayscale data. By incorporating both cues, the system provides more robust tracking, particularly in challenging conditions.

This feature introduces additional processing overhead; however, it can significantly improve track stability when grayscale-only tracking is insufficient. Enable color tracking in scenarios where maintaining a reliable track is difficult.

To enable color tracking, set *modeFlags bit 2* in the [SetTrackingParameters \(0x0C\)](#) message. To enable color tracking from Panel Plus, use the *Advanced Tracking* menu.

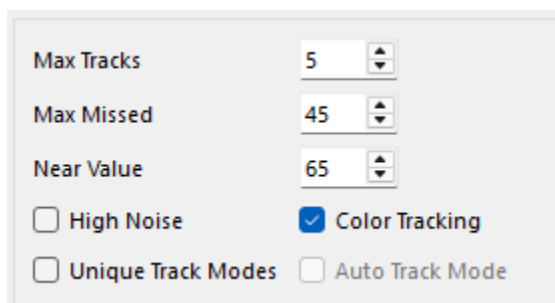


Figure 14: Color Tracking option

The example in [Figure 15](#) shows a scene where color tracking improves performance due to the strong color contrast between the boat and the ocean. Without color tracking, the tracker may struggle when waves and other scene elements interfere with traditional tracking methods. The distinct color difference between the boat and the water enables more reliable tracking.



Figure 15: Color Tracking Example



4.7 Reinitialize Primary

Software 2.22.xx and above.

If acquisition assist does not provide the optimal track box size, reinitializing the track is a low-risk way to update the track box size.

Sending the reinitialize command as part of the [ModifyTrackIndex \(0x17\)](#) message causes the acquisition assist logic to run again. A new track will start in the background. If there is high confidence that this track is better than the original, it will replace the original after ~1 second.

It is important to note that acquisition assist has limits and will not find an optimal track box size very different from the current size. In this case, adjusting the track box size is the best option. Adjusting the track box with acquisition assist enabled causes acquisition assist to run instantaneously instead of in the background.

4.8 Detection Track Cueing

Starting a track on a moving target can be difficult. Using any of the detection modes outlined in [EAN-Detection-Modes](#) can make this easier. Vehicle and Drone MTI modes were specifically designed to aid in starting tracks on moving objects, however the following logic could be used for any of the other modes as well. The following sections outline a couple of different ways to transition a detection to a primary track.

4.8.1 Mouse Click or Joystick

To turn a detection into a primary track, use the [ModifyTracking \(0x05\)](#) or [StartTracking \(0x08\)](#) command with a modify mode value of 5 or 6.

 *For this to work correctly it is important to setup a nearVal using the [SetTrackingParameters \(0x0C\)](#) command for your application.*

NearVal defines the area around an existing target that interactions can take place. For example, if using a joystick as a primary controller, set up the *Near Value* in Panel Plus and then create an overlay on the video with a circle in the middle of the screen with the radius equal to the *nearVal*. The example in [Figure 16](#) has a *Near Value* of 200 pixels.

To make a track primary when running *Vehicle MTI*, drive it near the center of the screen with the joystick. Click a button to send a message to start a track with the correct modify mode and coordinates at the center of the screen.

Alternately, this can work with a mouse or a touchscreen screen by simply clicking within the *nearVal* pixels of the detected target.

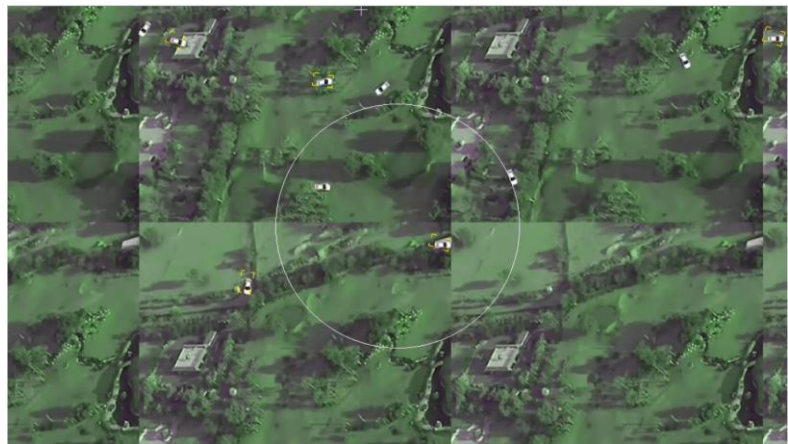


Figure 16: Starting a Track Using Near Value



4.8.2 Starting from PiP

Software 3.00.xx and above.

When using any of the Picture-in-Picture (PiP) modes in the drop-down menu, the location of the track in the PiP windows can be used to start a track on that target.

Use the [ModifyTrackIndex \(0x17\)](#) message and specify the PiP index in the message. With this feature a user interface can be designed that allows you to click on the PiP targets instead of clicking on the moving object to start a track.

Set up PiP in Panel Plus and then check the *Track from PiP* checkbox in the *Track* tab. In this mode, left-click on the PiP window to change the track to a primary target. The IDD [MultiDisplay \(0xA5\)](#) message describes setting up PiP track display with each PiP location having a specific index.



Figure 17: Start Track from PiP

4.8.3 Selected Track

Software 3.00.07 and above.

When using Vehicle or Drone detection mode use the [ShiftSelectedTrack \(0x33\)](#) and [DesignateSelectedTrackPrimary \(0x32\)](#) messages to toggle through the tracks and designate a track as primary.

This makes it easy for joystick users to allocate one button to toggle through tracks, and one button to start a track.

By default, selected tracks appear just like any other non-primary track, however they can be configured to have a unique overlay using the [SetOverlayMode \(0x06\)](#) message as shown in [Figure 18](#).

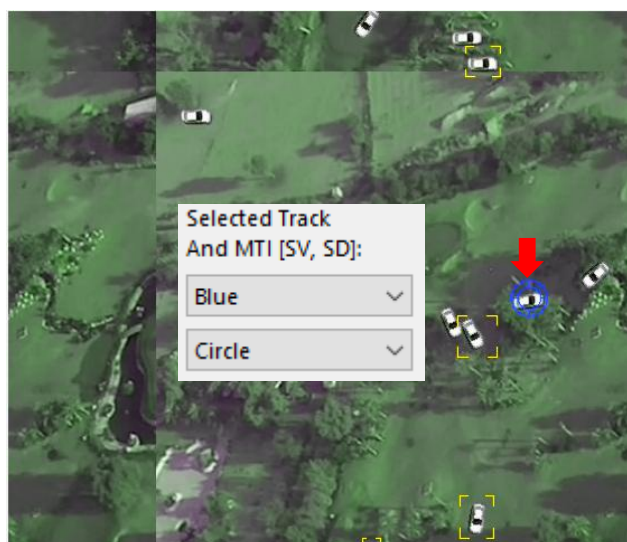


Figure 18: Designate Selected Track with Unique Overlay



4.9 Precision Acquisition

Software 3.8.x and above.

Precision Acquisition is in the *Advanced Tracking* dialog window (*Track tab* » *Advanced tracking settings*). It enables the user to initiate tracks on a past frame.

The system will then track through the queued frames until the present time is reached and then continue to track as normal.

When initiating a track using the *Precision Acquisition* feature, the amount of pause time is set to queue past frames to allow a track to start from a previous frame.

To enable this feature, the *Max Pause Time* must be greater than 0. This will enable the system to begin storing a history of frames in preparation for the *Precision Acquisition* track initiation.

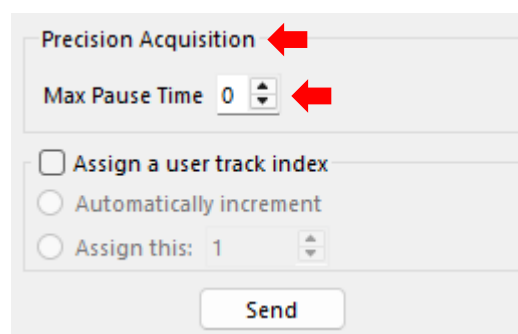


Figure 19: Advanced Tracking

4.9.1 Setting Up Max Pause Time

Software 3.8.x and above.

When enabling the *Precision Acquisition* feature, the time in seconds can be set in *Max Pause Time* in the [SetTrackingParameters \(0x0C\)](#) message or adjust *Max Pause Time* in the *Advanced Tracking* dialog window.

Max Pause Time is measured in seconds, up to a maximum of 20. This value may be reduced based on the number of active cameras, their frame rate, and resolution. A longer pause time allows the system to retain more history, which provides additional time to select a track from earlier frames. The total number of stored frames depends on camera frame rate and image size. The tables provide general guidelines rather than listing all possible configurations.

Table 3: 41xx-OEM Max Pause Time and Video Input Examples

Video Input	Number of Video Inputs			
	1	2	3	4
480p30	20	20	20 (10)	20 (10)
480p60	20	20	20 (10)	20 (10)
720p30	20	20	20 (10)	20 (10)
720p60	20	20	16 (10)	12 (10)
1080P30	20	20	14 (10)	10
1080p60	20 (10)	10 (5)	7 (5)	5

Table 4: 17xx-OEM/4000-OEM Max Pause Time and Video Input Examples

Video Input	Number of Video Inputs			
	1	2	3	4
480p30	20	20 (10)	20 (5)	20 (5)
480p60	20	20 (10)	20 (5)	18 (5)
720p30	20	20 (10)	16 (5)	12 (5)
720p60	20 (12)	12 (6)	8 (3)	6 (3)
1080P30	20 (10)	10 (5)	7 (2)	5 (2)
1080p60	10 (10)	5 (5)	3 (2)	2



Values in parentheses indicate limits prior to firmware version 3.11.x.

System memory is used to queue the frames. Consideration should be made if other programs are using system dynamic memory.

4.9.2 Start a Track Using Precision Acquisition

Once the *Max Pause Time* is set to a value greater than 0, the system will have previous frames available to initiate tracks.

To select one of these past frames to start a track, the MISB Precision Time Stamp that is associated with every network MISB frame can be used along with other parameters in the [StartTracking \(0x08\)](#) message to start the track.

When the system receives the message, if a corresponding MISB Precision Time Stamp is found in the queued history, the tracking system will start tracking on the associated frame and catch up to the current time continuing to track through each frame until the present time is reached.

4.9.3 Using Panel Plus

To use *Precision Acquisition* from Panel Plus, if the Max Pause Time was set up, the *Precision Acquisition* button can be used to pause the video frame and the associated MISB Precision Time Stamp. When the frame is paused, the *Precision Acquisition* button will turn red.

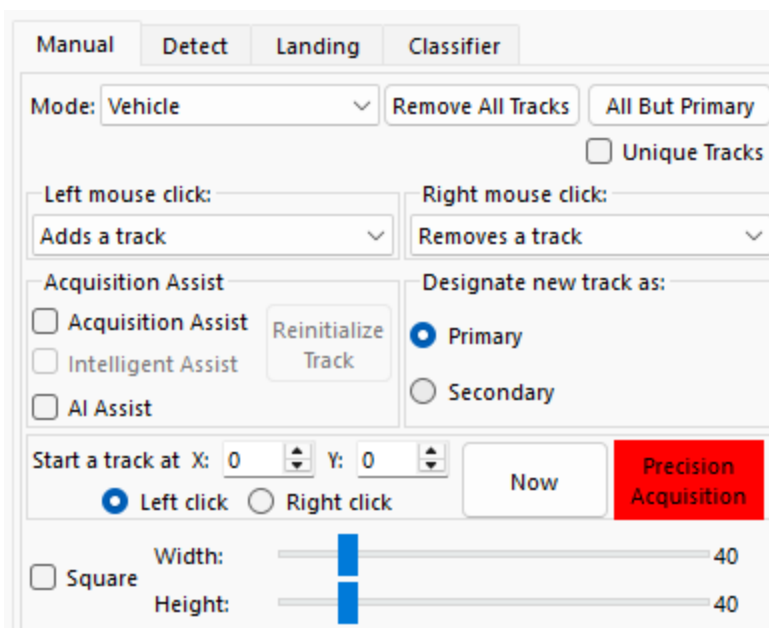


Figure 20: Track Tab, Manual Track Tab Precision Acquisition Button

Once the frame is paused, the mouse can be used to lasso a specific track box size. This allows a precision selection of the desired track target. Once the mouse is released, the system will resume the frame and initiate tracking as shown in [Figure 21](#).



Figure 21: Lasso Target on Paused Frame

When the system is catching up, the track box will draw a solid track box while catching up to the present time. Once the track has processed the history of frames and begins tracking at present time, the track box will transition to normal tracking.

 *Precision Acquisition uses the MISB Precision Time Stamps. The H264 and H265 MPEG2 transport streams are required to use this feature.*

5 Tracking Through Zoom and Rotation

The tracker works through slow changes in zoom and/or rotation. To track through faster changes the tracker must have an accurate estimate of the changes. Set the registration parameters *Max Zoom* and/or *Maximum Rotation* to non-zero values in the *Stabilization and Registration* dialog window (Panel Plus main menu » *Configure* » *Stabilization*), or use the [RegistrationParameters \(0x45\)](#) message.

A good starting point is 5. Increasing these values will increase the overall system processing load. Look at system performance in the *Performance Graph* before and after adjusting the parameters.

The tracker will often fail through large jumps in zoom. If it is not possible to reduce the step size or use a continuous zoom, re-designating the track after the zoom with the estimated size and location may be required for optimal performance.

5.1 Zoom Scaling

Software 3.6.x and above.

When tracking through zoom the underlying model will adjust its size with the registration zoom changes, but the reported and displayed track box will remain constant. To allow the track box size to change with the estimated zoom enable zoom scaling in the [SetTrackingParameters \(0x0C\)](#) message or check the *Scale target with zoom* box in the *Advanced Tracking* dialog window (*Track* tab » *Advanced tracking settings*).

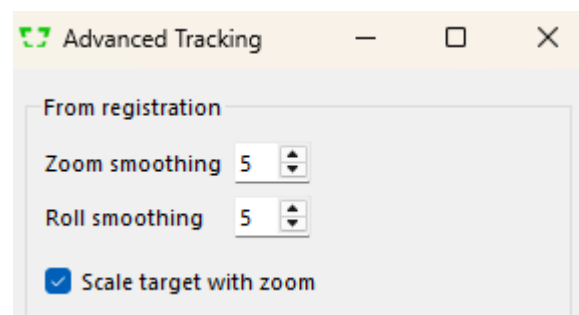


Figure 22: Advanced Tracking



5.2 Zoom and Roll Smoothing

Software 3.6.x and above.

By default, rotation and zoom values from registration are smoothed to prevent error buildup and improve tracking. The default smoothing value is 5 for both zoom and rotation (also known as roll). If necessary, the smoothing parameters can be adjusted to enhance tracking in certain cases in the *Advanced Tracking* dialog window.

Example use cases:

- For a smooth zoom-in or zoom-out, setting the *zoomSmoothing* parameter to 0 can lead to better results and a track that scales better with the zoom. This is also applicable when moving toward or away from the target instead of actively zooming the camera.
- When constantly orbiting a stationary target, reducing the *rollSmoothing* can result in a more stable track that is less likely to drift from the target. The reticle will not rotate in this case, but the underlying model will.

6 Tracking Through Occlusions

The ability to track through occlusions is a key feature of the tracker but there are a couple of things that can be done at a system level to improve performance.

6.1 Coasting State

It is important to understand when the tracker is in a coasting state. Look for the high bit of the track confidence to be set in the telemetry messages to indicate that the tracker is coasting. Using hard coded values is not enough since the confidence used internally may vary by algorithm.

6.2 Nudging / Resizing During Coast

While coasting it is important not to adjust the track box size. Adjusting the tracker box during coasting may cause the track to be lost because the tracker will refocus on the object that is occluding the target causing the track to be lost.

Nudging the track will have a similar effect in software version 2.25.xx and earlier. Starting in software version 3.00.xx, nudging during a coast will adjust the estimated target position but will not cause the tracker to re-train. If the tracker velocity estimate is off by a small trace and the track box is drifting from the target during a long occlusion, this can be a useful way to correct these errors. To do this it may be required to use larger nudge values than typically used during active tracking.

7 Additional Tracking Parameters

7.1 Max Tracks

Max Tracks refers to the maximum number of manual tracks and tracks generated from the Vehicle, Person, and Drone detection modes. In software version 3.7.x and below, the allowable range for the number of tracks is 1 to 5 on supported OEMs (1750-OEM, 4000-OEM, and 4100-OEM). In version 3.8.x and above, the allowable range increases to 1 to 10 with Full Tracking. Starting with version 3.9.x software, the maximum value increases to 20 tracks with Advanced Tracking on supported OEMs (17xx-OEM, 4000-OEM, and 4100-OEM).



 *The default value is 5 in all software versions.*

It is possible to adjust the *Max Tracks* value while tracks are active. However, any tracks with an index exceeding the new maximum will be automatically removed. If the track designated for removal is identified as the primary track, it will replace the track at index 0. This value can be modified with the [SetTrackingParameters \(0x0C\)](#) message.

7.2 Max Missed Frames

Max Missed Frames controls the maximum number of frames to keep searching for a track after the software stops discovering it. The default setting is 45, which corresponds to 1.5 seconds with 30 Hz video. Once the tracker has failed to find the track for *Max Missed Frames* it will drop the track.

Tuning this parameter correctly depends on the use case scenario. If it is important to know that the track is lost, and long occlusions are unlikely, either leave the number at the default setting or make it smaller. If longer occlusions are needed increase the number.

 *Improvements to velocity estimation in the tracker algorithm in 3.00.xx software help it to reacquire the track after a larger number of missed frames.*

7.3 NearVal

NearVal defines the radius of engagement used with the [ModifyTracking \(0x05\)](#) or [StartTracking \(0x08\)](#) message. Once the radius has been set, clicking anywhere within this range of an existing target designates it as primary, stop the track, etc.

A use-case example would be running Vehicle MTI and using a large *nearVal* to designate tracks near the center of the frame as primary. *Nearval* works in tandem with the modify mode parameter. See the [IDD](#) for specifics on the modify mode options. For more information on modify modes see the [Appendix C](#).

7.4 Rotate Nudge

Enable this option in the [NudgeTrackingCoordinate \(0x0A\)](#) message to send nudge commands in display coordinates instead of camera coordinates. For example, if a 180° rotation of the display is required because the camera is mounted upside down, this option ensures that the nudge commands are handled correctly.

7.5 Forced Coasting

Software 2.24.xx, 3.00.xx, and 3.4.x and above.

Forced Coasting was introduced in 2.24.xx software with the Freeze Updates mode. Freeze Search mode was added in 3.00.xx software. Freeze Propagation mode was added in software 3.4.x. These modes can be set or cleared from the [ModifyTrackIndex \(0x17\)](#) packet.

For more information about whether this feature can help with specific use case scenarios, send a video to Sightline [Support](#) for review and recommendations.

7.5.1 Freeze Updates

The software searches for the object in every frame, updates its position and momentum, etc., but the model will not update as new data is obtained. In most cases using this feature will decrease tracking performance because model updates are required to maintain persistent tracks.



However, there are cases where temporarily stopping the updates can be beneficial (~1 sec or less). For example, if there is a cloud of smoke between the camera and the target or some other known disturbance.

It can also be helpful to enter this mode prior to the target being partially occluded. This is not a recommended normal mode of operation to deal with full occlusion. The tracker is already designed to manage this type of situation.

7.5.2 Freeze Search

The software will not search for the target or update any target state in this mode. The reported position in this mode will be the estimated position based on the current target velocity when this state was entered and the frame-to-frame registration values. If the number of frames since entering this mode reaches the *Max Missed Frames* value the mode will be exited automatically and the tracker will search for the target, if it fails to find it the track will drop at this point.

7.5.3 Freeze Propagation

The software will continue to search for the target and update its position and model, but the velocity, momentum, and registration information will not be used in deciding where to search in the image.

This mode may be helpful if the entire image momentarily becomes undiscernible and if the target being tracked will not move significantly during the time it cannot be seen.

 *In Panel Plus this is also referred to as Freeze Prop.*

7.6 Transition Between Person and Vehicle Mode

Software 3.2.x and above.

When switching tracking modes, the track that is active is reset. In most situations this is acceptable behavior. However, in some situations it may be desirable for the track state to persist when switching between Vehicle and Person modes. To persist the track state when switching between these modes, see the [SetTrackingParameters \(0x0C\)](#) message.

8 Tracker-Only Configurations

The tracker-only configuration provides a solution for customers who need an object tracking algorithm and telemetry data, but do not need another video processing functionality.

Tracker-only configurations use the same tracking algorithms as the full featured SLA configuration. The tracker-only option is outlined in the [Appendix B](#). Supported hardware platforms can be easily upgraded to full SLA functionality.



9 Command and Control

The following packets are used to set up and control the tracking algorithms. Details can be found in the web based [IDD](#).

 A downloadable PDF version is also available on the [Software Releases](#) page.

9.1 IDD Menu

Within the IDD the main menu references Modules (a category sort of commands and data) and Classes (packets, structs, unions and interfaces in the form of SLA[*name of class*]_t).

9.2 Reporting Results

Telemetry data related to tracking.

Message Name (Class)	Message ID	Module
TrackingPosition	(0x43)	Telemetry
TrackingPositions	(0x51)	Telemetry
TrackingPositionsExtended	(0xA0)	Telemetry
CoordinateReportingMode	(0x0B)	Telemetry
SetTelemetryDestination	(0x64)	Telemetry

9.3 Display Related Commands

Message Name (Class)	Message ID	Module
Set Overlay Mode	(0x06)	Overlays

9.4 Tracking Related Commands

Message Name (Class)	Message ID	Module
ModifyTracking	(0x05)	Tracking
StartTracking	(0x08)	Tracking
ModifyTrackIndex	(0x17)	Tracking
StopTracking	(0x09)	Tracking
SetTrackingParameters	(0x0C)	Tracking
NudgeTrackingCoordinates	(0x0A)	Tracking
ShiftSelectedTrack	(0x33)	Tracking
DesignateSelecte TrackPrimary	(0x32)	Tracking
StopSelectedTrack	(0x3C)	Tracking
RegistrationParameters	(0x9E)	Stabilization



10 Troubleshooting

Question	Answer
It is too complicated to always switch tracking modes. Which tracking mode would work well for most tracking scenarios?	Although tracking mode is important, using Vehicle mode tracks moving or stationary targets in many different scenarios and is a good choice for many tracking applications. Choosing stationary mode would result in poor tracking performance on moving objects
What is the difference between scene tracking mode and using the registration results to implement a scene tracker?	The scene tracking mode is a simple implementation of a scene tracker that takes the registration information and outputs the results in the TrackingPosition (0x43) packet in camera coordinates. Using the registration results directly still requires looking at the TrackingPosition (0x43) packet, but uses the scene translations as opposed to the tracking row and col. The Tracking Position packet contains additional fractional information about scene translation. See the IDD on TrackingPosition (0x43) for an example of generating floating point scene translations using sceneCol, sceneRow, sceneColFrac8, sceneRowFrac8. The benefit to using the registration results directly is having more control managing feed-forward user commands. For example, driving the scene to the right in <i>Scene Tracking</i> mode using the nudge commands will result in a jumpy move. Using the registration results directly makes it easier to incorporate user commands in the control loop to provide smoother gimbal control.
How can a track box be set up with a different width and height?	This is supported in 2.24.xx. In previous versions use <i>Acquisition Assist</i> or <i>Vehicle MTI</i> to adjust the track box.
When using intelligent assist is the <i>Near Value</i> important?	No, <i>Near Value</i> has no relationship with intelligent assist. Intelligent assist is designed to optimize the track box size under a specific set of conditions after tracking has been active for a while. <i>Near Value</i> starts new tracks or transitioning a track from a detection to a primary track.
What is the minimum size object that can be tracked?	Tracking has been shown to work with as few as 4 pixels (2x2 box). For more robust tracking, track objects that are at least 100 pixels (10x10 box).

10.1 Questions and Additional Support

For questions and additional support, please contact [Sightline Support](#). Additional support documentation and Engineering Application Notes (EANs) can be found on the [Product Support Documentation](#) page in the [Sightline Knowledge Base](#).



Appendix A - Tracker Performance Summary

Criteria	Expected Performance
Minimum target size (pixel)	2x2 pixels. Robust tracking at 10x10 pixels.
Minimum target contrast with respect to background	1% (tested in a simulated environment).
Frame rate / Telemetry output rate	Typically used at 30Hz. 60 Hz possible on the 4000-OEM, 4100-OEM, and 3000-OEM at some frame sizes. Very slow frame rates (e.g., 9Hz) may lead to reduced tracking performance.
Tracking Gate Position modes	Manually movable over the entire FOV. Automatically movable while tracking a target within the FOV.
Handling of lost track	During obscuration, maximum coasting during hidden target is 8.5 seconds. In real world examples, 4-5 seconds is typical.
Tracking accuracy	The center of the track box is reported with Subpixel resolution.
Minimum size of the tracker window	Minimum track box size (not target size) is 4x4 pixels. Using intelligent assist mode, the tracker window automatically changes size to maintain an optimal ratio with respect to the changing target size. See the Intelligent Assist section for more information.
Maximum size of the tracker window	500x500 pixels.
Maximum tracking velocity	Set by registration change maximums. ¼ frame-per-frame is a safe estimate of max velocity. Once tracking begins and the gimbal is being slewed by the operator, the frame-to-frame shift in pixel space is typically much smaller because the system works to keep the camera centered on the tracked object.
Maximum track acceleration	20 pixels/frame for Vehicle mode. 40 pixels/frame for Drone mode.



Appendix B - Tracking Only Comparison

Table B1: Tracking-Only and SLA Comparison

Attribute	Full SLA Tracking Advanced + AI Assist (3.10 and above)	Full SLA Tracking Advanced (3.9 and above)	Full SLA Tracking Advanced (3.8 and above)	Full SLA Tracking (3.8 and above)	Full SLA	Tracker Only
Tracker modes supported	Vehicle Stationary Scene Static Drone Person	Vehicle Stationary Scene Static Drone Person	Vehicle Stationary Scene No Registration Drone Person	Vehicle Stationary Scene No Registration Drone Person	Vehicle Stationary Scene No Registration Drone Person	Vehicle Stationary No Registration Drone Person
Precision Acquisition (track to the future)	✓	✓	✓	NA	NA	NA
Maximum number of simultaneous tracks	20	20	10	10	5	1
Frame size processed	HD or SD	HD or SD	HD or SD	HD or SD	HD or SD	SD
Track box size adjust	✓	✓	✓	✓	✓	✓
Track box nudging	✓	✓	✓	✓	✓	✓
Track through zoom and rotation	✓	✓	✓	✓	✓	✓
Acquisition Assist	✓	✓	✓	✓	✓	✓
Intelligent Assist	✓	✓	✓	✓	✓	✓
AI Track Assist	✓	NA	NA	NA	NA	NA
Scene Tracking (aka Scene Steering and Scene Lock)	✓	✓	✓	✓	✓	NA
Detection Track Cueing	✓	✓	✓	✓	✓	NA
Hardware platforms supported	17xx, 4100, 4000	17xx, 4100, 4000	1750, 4100, 4000	1750, 4100, 4000	1750, 4100, 4000, legacy 1500, 3000	1750, 1700, 1500
Other video processing functions that can be added to the unit.	All Sightline functions can be added.	All Sightline functions can be added.	All Sightline functions can be added.	All Sightline functions can be added.	All Sightline functions can be added.	None. Functions are limited to tracking only. Units can be updated to full performance through a license file purchase.



Appendix C- Modify Modes

This section explains the relationship between the modify mode values and the [ModifyTracking \(0x05\)](#) or [StartTracking \(0x08\)](#) messages. Refer to the [IDD](#) to help choose the one that is suitable for your application.

C1 Modify Modes

Table B1 shows how each modify mode can be implemented through a different combination of settings in Panel Plus.

Table C1: Modify Modes

Modify Mode Value	Panel Plus Tab	Panel Plus Controls
0 - Show cursor only. If using this mode, display coordinate and source coordinate bits are ignored.	Overlay	
1 - Kill any existing targets and then designate a new primary target at the cursor.	Tracking	
2 - Designate a primary target at the cursor.	Tracking	
3 - If there is a track near the coordinates, move track to coordinates and make secondary.	Tracking	
4 - If there is a track near the coordinates, move track to coordinates and make secondary. Otherwise, add a new secondary track at coordinate location.	Tracking	



(Modify Modes table continued)

Modify Mode Value	Panel Plus Tab	Panel Plus Controls
5 - Designate track near coordinates as primary target. Also supports starting primary tracks from SA or ST MTI tracks.	Tracking	Left mouse click: Move nearest or make new ☐ Acquisition Assist ☐ Intelligent Assist ☐ AI Assist Reinitialize Track Right mouse click: Re-Designate existing nearest Designate new track as: ☒ Primary ☐ Secondary
6 - Designate track near coordinates as primary target if there is one. If not, add new track at coordinates and make primary. Also supports starting primary tracks from SA or ST MTI tracks.	Tracking	Left mouse click: Adds a track ☐ Acquisition Assist ☐ Intelligent Assist ☐ AI Assist Reinitialize Track Right mouse click: Re-designate or make new Designate new track as: ☒ Primary ☐ Secondary
7 - If there is a track near coordinates, move track to location of coordinates and designate as primary. Otherwise, add a new track at coordinates and make primary.	Tracking	Left mouse click: Move nearest or make new ☐ Acquisition Assist ☐ Intelligent Assist ☐ AI Assist Reinitialize Track Right mouse click: Re-Designate existing nearest Designate new track as: ☒ Primary ☐ Secondary
8 - If there is a track near coordinates, move track to location of coordinates and designate as primary. Otherwise, kill all existing tracks and add a new primary track at location of coordinates.	Tracking	Left mouse click: Move nearest or make new ☐ Acquisition Assist ☐ Intelligent Assist ☐ AI Assist Reinitialize Track Right mouse click: Mode 8 Designate new track as: ☒ Primary ☐ Secondary
9 - Kill track near coordinates	Tracking	Left mouse click: One track only ☐ Acquisition Assist ☐ Intelligent Assist ☐ AI Assist Reinitialize Track Right mouse click: Removes a track Designate new track as: ☒ Primary ☐ Secondary



(Modify Modes table continued)

Modify Mode Value	Panel Plus Tab	Panel Plus Controls
10 - Kill all tracks but the primary track.	Tracking	
11 - Reserved	N/A	N/A
12 - Designate track near coordinates as secondary target. Also support starting secondary tracks from SA or ST MTI tracks.	Tracking	
13 - Designate track near coordinates as secondary target if there is one. If not, add new track at coordinates and make secondary. Also support starting secondary tracks from SA or ST MTI tracks.	Tracking	

C2 Modify Modes - Software Version 2.25.xx and Earlier

Panel Plus support for the various modify modes was limited to software 2.25.xx and earlier versions. The modify modes shown in this table cover the most common use cases. To implement other modes requires user code or custom macros/send commands.

Table C2: Modify Modes - Software Versions 2.25.xx and Earlier

Modify Mode Value	Panel Plus Tab	Panel Plus Controls
0 - Show Cursor only. If in this mode Display Coordinate and Source Coordinate bits are ignores.	Overlay	
1 - Kill any existing targets and then designate a new primary target at the cursor.	NA	Not supported in this version of Panel Plus.

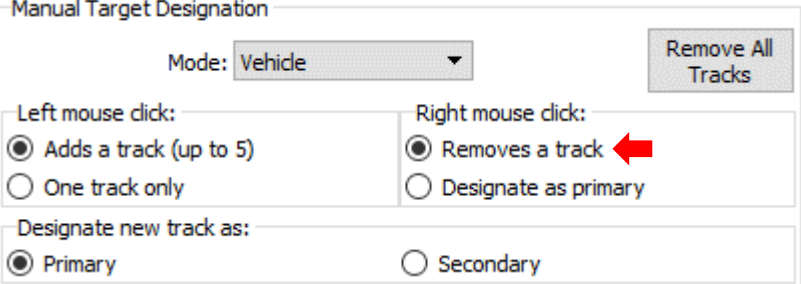


(Modify Modes - Software Versions 2.25.xx and Earlier table continued)

Modify Mode Value	Panel Plus Tab	Panel Plus Controls
2 - Designate a primary target at the cursor.	NA	Not supported in this version of Panel Plus.
3 - If there is a track near the coordinates, move track to coordinates and make secondary.	NA	Not supported in this version of Panel Plus.
4 - If there is a track near the coordinates, move track to coordinates and make secondary. Otherwise, add a new secondary track at coordinate location.	Tracking	
5 - Designate track near coordinates as primary target. Also supports starting primary tracks from SA or ST MTI tracks.	NA	Not supported in this version of Panel Plus.
6 - Designate track near coordinates as primary target if there is one. If not, add new track at coordinates and make primary. Also supports starting primary tracks from SA or ST MTI tracks.	Tracking	
7 - If there is a track near coordinates, move track to location of coordinates and designate as primary. Otherwise, add a new track at coordinates and make primary.	Tracking	



(Modify Modes Software Versions 2.25.xx and Earlier table continued)

Modify Mode Value	Panel Plus Tab	Panel Plus Controls
8 - If there is a track near coordinates, move track to location of coordinates and designate as primary. Otherwise, kill all existing tracks and add a new primary track at location of coordinates.	NA	Not supported in this version of Panel Plus.
9 - Kill track near coordinates	Tracking	
10 - Kill all tracks but the primary track.	NA	Not supported in this version of Panel Plus.
11 - Reserved	NA	N/A
12 - Designate track near coordinates as secondary target. Also support starting secondary tracks from SA or ST MTI tracks.	NA	Not supported in this version of Panel Plus.
13 - Designate track near coordinates as secondary target if there is one. If not, add new track at coordinates and make secondary. Also support starting secondary tracks from SA or ST MTI tracks.	NA	Not supported in this version of Panel Plus.